

Queuing Theory: Contributions and Applications in the Field of Health Service Management – A Bibliometric Approach

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Abstract: This article aims to analyze contributions and applications of Queuing Theory (QT) in solving health service management problems. A literature bibliometric review was carried out to present a chronological classification of health care systems examined with the aid of queuing models with the purpose of describing the main trends in QT application to assist decision makers in health services. Methodology: A bibliometric review was carried out to investigate the area of health supported by queuing models. Articles describing models of problems were sought and their topics or keywords related to QT and health care or health problems of the population. Results: 314 articles addressing the applications of QT in Health Care Management (HCM) from 2014 and 2020 were analyzed. Conclusion: It was found that QT contributes to improve health services and provides managers with resources to achieve such improvement. It was also evidenced that it is widely applied in emergency services.

Keywords: Queue Theory, Simulation, Bibliometric Review, Fast Track, Lean Healthcare.

1. INTRODUCTION

The current scenario of organizations requires a permanent effort so that they can remain competitive. According to Schmidt and Cohem (2014), the digital revolution influencing individuals, organizations, communities and federal governments has led to a significant paradigm shift. It is difficult to predict the characteristics of modern society, and emerging cultural, political, and commercial systems are going to be affected by globalization in distinct regions of the globe.

Some concerns as international competitiveness, application and use of emerging technologies, sustainability and development, organizational learning and ethical business challenges comprise a volatile and highly complex context for taking actions on innovation and permanence (BHAGAT; MCDEVITT; BALIGA, 2017; NICOLAIDES, 2017). In contrast, this context provides those who surviving in this competitive environment with several improvement opportunities. One of the main challenges to be faced in order to succeed under these conditions lies in existing gains involved in relationships between quality, innovation, competitiveness and performance (AL-SHUAIBI; ZAIN; KASSIM, 2016). In a healthcare environment, challenges particularly arising from a management perspective are also associated with common healthcare issues at times (JORMA et al., 2016). Vile et al (2016) addresses capacity planning as a primary issue through the development of a workforce planning tool integrating forecast, priority queuing theory and

programming methods in a decision support system in order to optimize resource allocation in terms of capacity. They pose managing and planning hospital capacity as one of the great challenges to be faced in healthcare services. Healthcare organizations are characterized as complex institutions, and such complexity derives from its ambiguous objectives, in addition to a qualitative character of its activities, use of multiple and complex technologies, shared power and plurality of professionals involved in their activities. Chaves et al (2021) conducted studies on the Fast Track implementation in healthcare units with the purpose to improve capacity in the flow of patients and speed up the fulfillment of relevant activities.

However, in the area of healthcare, it is the set of processes affecting the healing of a patient suffering from an affliction (GRABAN, 2012). In Santos et al (2021), the authors discuss a method to measure a company's maturity level by providing information on its features aiming to constitute continuous improvement, which might compel institutions to search for techniques like QT. The use of a diagnostic report tool might be encouraged, however, Borges et al (2021) regards understanding and implementing routine practices (katas) as important to maintain sustainability

Therefore, the objectives of this study are to analyze contributions and applications of QT in the field of problem management in health services through a bibliometric review in order to propose a system of chronological classification of healthcare with the aid of models to assess healthcare queues.

2. METHOD

PubMed, Scopus and Web of Science databases were used as sources to search articles published in the period of 2014 and 4/18/2020. The search strategy was divided into two stages. The first one consisted in identifying keywords or in their titles and abstracts the expression: "queuing theory" (Chart 1). The second one was aimed to define the inclusion criteria, which consisted of: (1) if the article describes a queuing model and its applications in the management of health services; and (2) if there is a clear association with typical health service management issues.

Table 1. Search criteria and selection of articles using the queuing theory in health services carried out on the Web of Science, PubMed and Scopus databases

Search Source	Web Of Science	PubMed	Scopus
Search date	04/15/2020	04/16/2020	04/18/2020
Terms	("queuing theory")	("queuing theory")	("queuing theory")
Criteria (filters)	(HEALTH CARE SCIENCES SERVICES); (HEALTH POLICY SERVICES); (MEDICAL INFORMATICS); (EMERGENCY MEDICINE); (MEDICINE RESEARCH EXPERIMENTAL)	NO FILTERS	(MEDICINE)
Articles collected	39	105	170

After collecting data on the initial set of articles, they were classified according to their proposition, decisions made about their application area, professionals involved in their application areas, desired results and how the QT was applied in studies. Reviewed articles are either directly or indirectly related to research on QT applications in the area of healthcare.

14 articles were selected out of the 314 articles sought. An intentional non-probabilistic sampling technique was used based on research discussing aspects of QT application in the

field of health service management problems. A convenience sampling approach was opted to control the influence of other unrelated factors and define the scope for selecting studies on QT and its applications. The PRISMA Flowchart - Main Items to Report on Systematic Reviews and Meta-analyses (GALVÃO, 2015) were used, which enabled to elaborate the Flowchart of Reviewed Literature and the identification of

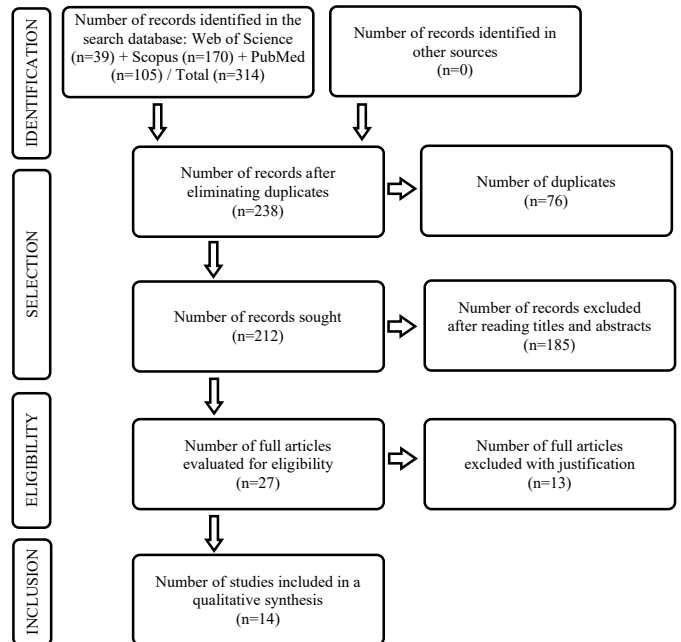


Figure 1. Flowchart of the research method used for selecting articles about QT and its applications in the area of healthcare. 14 articles were selected according to these parameters.

articles.

Tavares et al., (2018) define non-probabilistic sampling as being based on reasoning and depending exclusively on the researchers' criteria to delimit and form the sample. A convenience sampling approach was chosen to control the influence of other unrelated factors and define the scope for selecting studies on QT and its applications. The PRISMA Flowchart - Main Items to Report for Systematic Reviews and Meta-analyses (GALVÃO, 2015) was used, which allowed elaborating the Flowchart of Reviewed Literature and the identification, of initially 314 records out of which 14 were selected, according to Figure 1.

3. RESULTS AND DISCUSSION

3.1 Co-occurrence in titles and abstracts

A co-occurrence of terms was found in titles and abstracts of sought articles. The bibliometric review showed the presence of only 1 cluster, 33 items and 528 links, which reveals the presence of 3 main terms: Patient Flow, Simulation and Approach and 30 secondary terms. Based on the detection and frequency at which they appear in reviewed article, the diagram on Figure 2 was drawn. Large value-added content was disregarded due to proximity/distancing from main terms.

3.2 Keyword co-occurrence

A co-occurrence of keywords was found in reviewed articles. The review revealed the presence of 12 clusters that, for identification purposes, are represented using different colors with 1 item in each cluster, and this configuration is characterized by the absence of main keywords and a dispersion representing a low frequency of terms in the set of literature references being studied. In this case, keywords express concepts contained in works, and the terms found are keywords used in indexing to indicate an ordered collection of information and knowledge. In co-occurrence, a simultaneous occurrence indicates a close relationship between them.

3.2 Co-occurrence of authorship

Co-authorship was found in studied articles. The bibliometric review identified the presence of 14 clusters consisting of 48 authors. Figure 4 represents co-authorship. These clusters have no connection with each other due to the absence of main authors in the set of publications found, in addition to the fact that there is no citation relationship between authors, thus demonstrating that such dispersion corresponds to a diversity of authors who have seldom published in the same area of knowledge.

Among the selected articles addressing the QT, some focusing on the use of QT in health services problem management are going to be highlighted and analyzed.

Cildo, Ibarra and Mallor (2019) used the QT through a extensive computational analysis in order to show the flexibility of queue theory called accumulating priority queue with a finite horizon denoted by (APQ-h) and the accumulating priority queue (APQ), as well as its superiority over other disciplines of sheer priority in a real-world and a variety of emergency unit scenarios. Gkika et al (2019) sought to apply QT to estimate the impact of a rapid diagnosis on the average length of stay and the use of health resources.

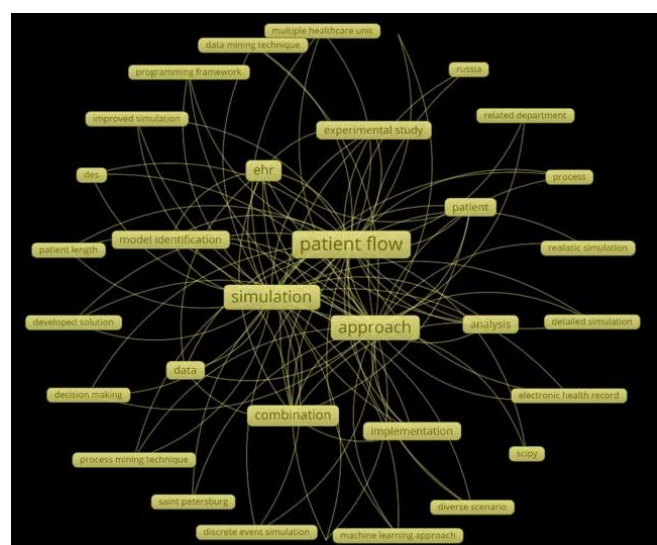


Figure 2. Co-occurrence of terms in titles and abstracts of 14 selected articles.

Varney, Bean and Mackay (2019) studied the link between patient admission and discharge behaviors, among other potential occupation management behaviors, so as to make a correlation between the moment of arrival and the length of stay using the queuing theory as a simple and robust method to assess correlations and explain their structure while simulating the occupation of an ICU. Myszewski and Sinha (2018) used QT as a methodology to describe phenomena in which not all planned tasks are performed within recommended specification limits.

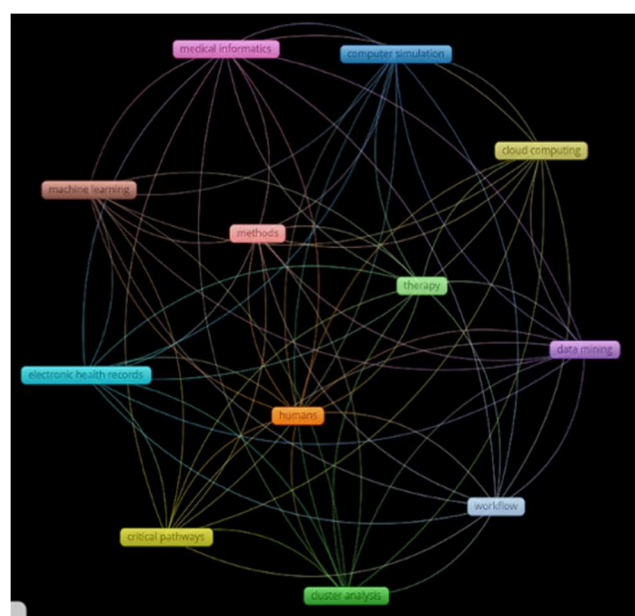


Figure 3. Co-occurrence of terms found in keywords of 14 selected articles, each being identified using a different color
Source: VOSVIEWER.

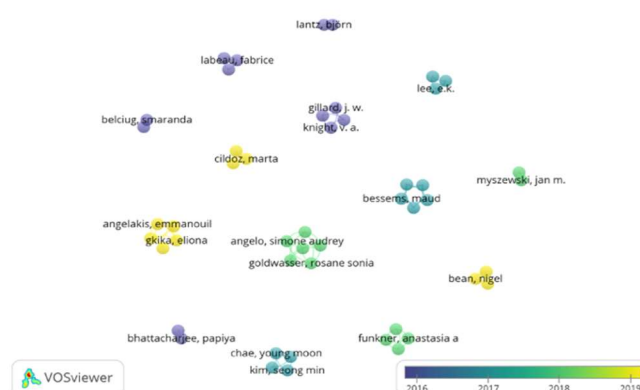


Figure 4. Co-occurrence of authors and their interrelationships found in 14 selected articles. Different colors are related to the year of publication shown in the timeline.

According to Lin, Patrick and Labeau (2014) QT was used to study the development of a model with two connected queues: one regarding patient flow to access the Emergency Department and another with respect to patient access to the Hospital Unit. Based on this flow model, QT was used to estimate the average waiting time of patients. Using specific and priority waiting time targets, the necessary number of emergency resources and inmate unit was also estimated.

4. CONCLUSION

This article reviews and categorizes the applications of queuing theory in the modeling of hospital processes. As health services are directly related to safeguarding human lives, improving their system performance is a mandatory goal for these services and analyzed articles clearly show that QT can contribute to achieve such an improvement.

This fact can be identified in analyzed articles, especially those demonstrating that the QT increases the use of resources, decreases waiting time of patients, improves the productivity of systems and reduces waste when properly applied. It can also be concluded that QT is an effective and powerful modeling technique allowing managers to achieve their goals in improving the system. Another finding through this literature review is that QT can be applied with moderate ease in different hospital sectors, thereby providing good and quick estimates about the system performance. The selected articles also reveal that QT has been applied more frequently in emergency sectors by healthcare professionals.

Acknowledgements: The authors would like to thank the Brazilian Ministry of Health, UFF - Federal Fluminense University, and Euclides da Cunha Foundation. This Research is part of the "Lean Project in UPA 24h" that has been funded by the Brazilian Ministry of Health (TED 15/2021, number: 25000.103452/2021-14).

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